



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Spring 2024-25

Section: M

Lab Work (MID + FINAL)

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Submitted By

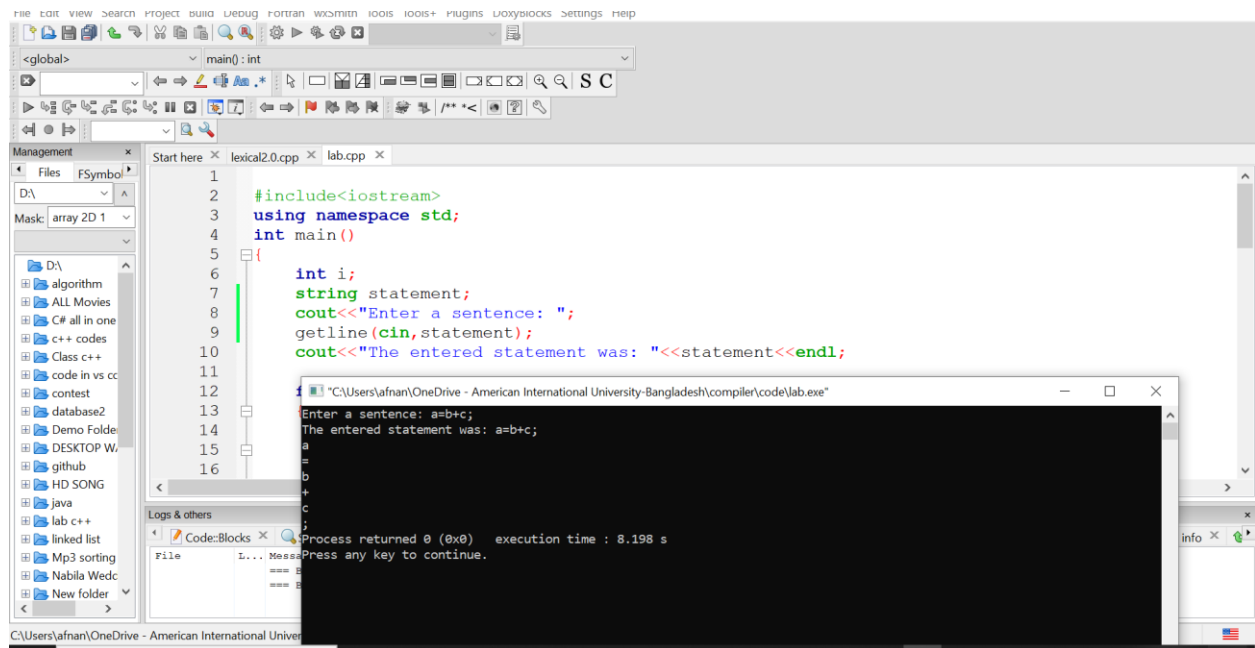
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Lexical Analyzer Code Skeleton version 1.0

Code:

```
1
2 #include<iostream>
3 using namespace std;
4 int main()
5 {
6     int i;
7     string statement;
8     cout<<"Enter a sentence: ";
9     getline(cin, statement);
10    cout<<"The entered statement was: "<<statement<<endl;
11
12    for(i = 0; i<statement.length(); i++)
13    {
14        if(statement[i] == '=')
15        {
16            cout<<endl;
17            break;
18        }
19        cout<<statement[i];
20    }
21
22    for(i = i; i<statement.length(); i++)
23    {
24        cout<<statement[i]<<endl;
25        i = i+1;
26        break;
27    }
28
29    for(i = i; i<statement.length(); i++)
30    {
31        if(statement[i] == '+')
32        {
33            cout<<endl;
34            break;
35        }
36        cout<<statement[i];
37    }
38
39    for(i = i; i<statement.length(); i++)
40    {
41        cout<<statement[i]<<endl;
42        i = i+1;
43        break;
44    }
45
46    for(i = i; i<statement.length(); i++)
47    {
48        if(statement[i] == ';')
49        {
50            cout<<endl;
51            break;
52        }
53        cout<<statement[i];
54    }
55
56    for(i = i; i<statement.length(); i++)
57    {
58        cout<<statement[i];
59        break;
60    }
61
62    return 0;
63 }
64
65
```

Output:



The screenshot shows a C++ IDE with the following components:

- File Explorer:** Displays a directory tree on the left with folders like 'algorithm', 'ALL Movies', 'C# all in one', 'c++ codes', 'Class c++', 'code in vs cc', 'contest', 'database2', 'Demo Folder', 'DESKTOP W...', 'github', 'HD SONG', 'java', 'lab c++', 'linked list', 'Mp3 sorting', 'Nabila Wedc', and 'New folder'.
- Code Editor:** Contains the following C++ code:

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int i;
6     string statement;
7     cout<<"Enter a sentence: ";
8     getline(cin,statement);
9     cout<<"The entered statement was: "<<statement<<endl;
```

- Output Window:** Displays the program's output:

```
Enter a sentence: a=b+c;
The entered statement was: a=b+c;
```

- Logs and others:** Shows the process return code and execution time:

```
Process returned 0 (0x0)   execution time : 8.198 s
Press any key to continue.
```

Lexical Analyzer Code Skeleton version 2.0

Code:

```
1
2 #include <iostream>
3 using namespace std;
4
5 int main ()
6 {
7
8     int i, temp;
9     int location_counter = 1;
10    string statement;
11
12    cout << "Enter the statement:";
13    getline (cin, statement);
14    cout << "The entered statement was:" << statement << endl;
15
16    if (
17        statement[0] == '0' || statement[0] == '1' || statement[0] == '2' || statement[0] == '3' ||
18        statement[0] == '4' || statement[0] == '5' || statement[0] == '6' || statement[0] == '7' ||
19        statement[0] == '8' || statement[0] == '9')
20    {
21        cout << "lexical error : " << statement[0] << " at location 0" << endl;
22        return 0;
23    }
24
25    else
26        cout << "<id," << location_counter << ">" << endl;
27
28    for (i = 1; i < statement.length (); i++)
29    {
30        if (statement[i] == '=')
31        {
32            break;
33        }
34    }
35
36    cout << "<=" << endl;
37    temp = i;
38
39    if (
40        statement[temp + 1] == '0' || statement[temp + 1] == '1' || statement[temp + 1] == '2' || statement[temp + 1] == '3' ||
41        statement[temp + 1] == '4' || statement[temp + 1] == '5' || statement[temp + 1] == '6' || statement[temp + 1] == '7' ||
42        statement[temp + 1] == '8' || statement[temp + 1] == '9')
43    {
44        cout << "lexical error : " << statement[temp + 1] << " at location " << temp + 1 << endl;
45        return 0;
46    }
47
48    else
49    {
50        location_counter++;
51        cout << "<id," << location_counter << ">" << endl;
52    }
53
54    for (i = temp + 2; i < statement.length (); i++)
55    {
56        if (statement[i] == '+')
57        {
58            break;
59        }
60    }
61
62    cout << "<+>" << endl;
63    temp = i;
64
65    if (
66        statement[temp + 1] == '0' || statement[temp + 1] == '1' || statement[temp + 1] == '2' || statement[temp + 1] == '3' ||
67        statement[temp + 1] == '4' || statement[temp + 1] == '5' || statement[temp + 1] == '6' || statement[temp + 1] == '7' ||
68        statement[temp + 1] == '8' || statement[temp + 1] == '9')
69    {
70        cout << "lexical error : " << statement[temp + 1] << " at location " << temp + 1 << endl;
71        return 0;
72    }
73
74    else
75    {
76        location_counter++;
77        cout << "<id," << location_counter << ">" << endl;
78    }
79
80    for (i = temp + 2; i < statement.length (); i++)
81    {
82        if (statement[i] == ';')
83        {
84            break;
85        }
86    }
87
88    cout << "<.>" << endl;
89
90    return 0;
91
92 }
93
```

Output:

```
lexical2.0.cpp - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global>
[Icons] [S] [C]

Management
Files FSymbol
D:\
Mask: array 2D 1
D:\
algorithm
ALL Movies
C# all in one
c++ codes
Class c++
code in vs cc
contest
database2
Demo Folde
DESKTOP W
github
HD SONG
java
lab c++
linked list
Mp3 sorting
Nabila Wedc
New folder

Start here lexical2.0.cpp lab.cpp
19 statement[0] == '0' || statement[0] == '9')
20 {
21     cout << "lexical error : '" << statement[0] << "' at location 0" << endl;
22     return 0;
23 }
24
25 else
26     cout << "<id," << location_counter << ">" << endl;
27
28
29 for (i = 1; i < statement.length (); i++)
30 {
31     if (statement[i] == '=')
32     {
33         *C:\Users\afnan\OneDrive - American International University-Bangladesh\compiler\code\lexical2.0.exe"
34     }
35     )Enter the statement:a=b+c
36     The entered statement was:a=b+c
37     <id,1>
38     <=>
39     <id,2>
40     <=>
41     <id,3>
42     <=>
43 }

Logs and others
Code::Blocks Process returned 0 (0x0) execution time : 6.960 s
File L... Mess Press any key to continue.
```